

Knowledge Management with (Bio)Croissant*

Steffen Vogler :: BioCypher Workshop :: 15 June 2026

* ... *BioCroissant* is the name of the working group, *Croissant* is the name of the exchange format

Data is Source Code for AI

- **Croissant makes datasets machine-readable without changing the data**

- One metadata layer across storage locations
- Compatible for Agentic tools

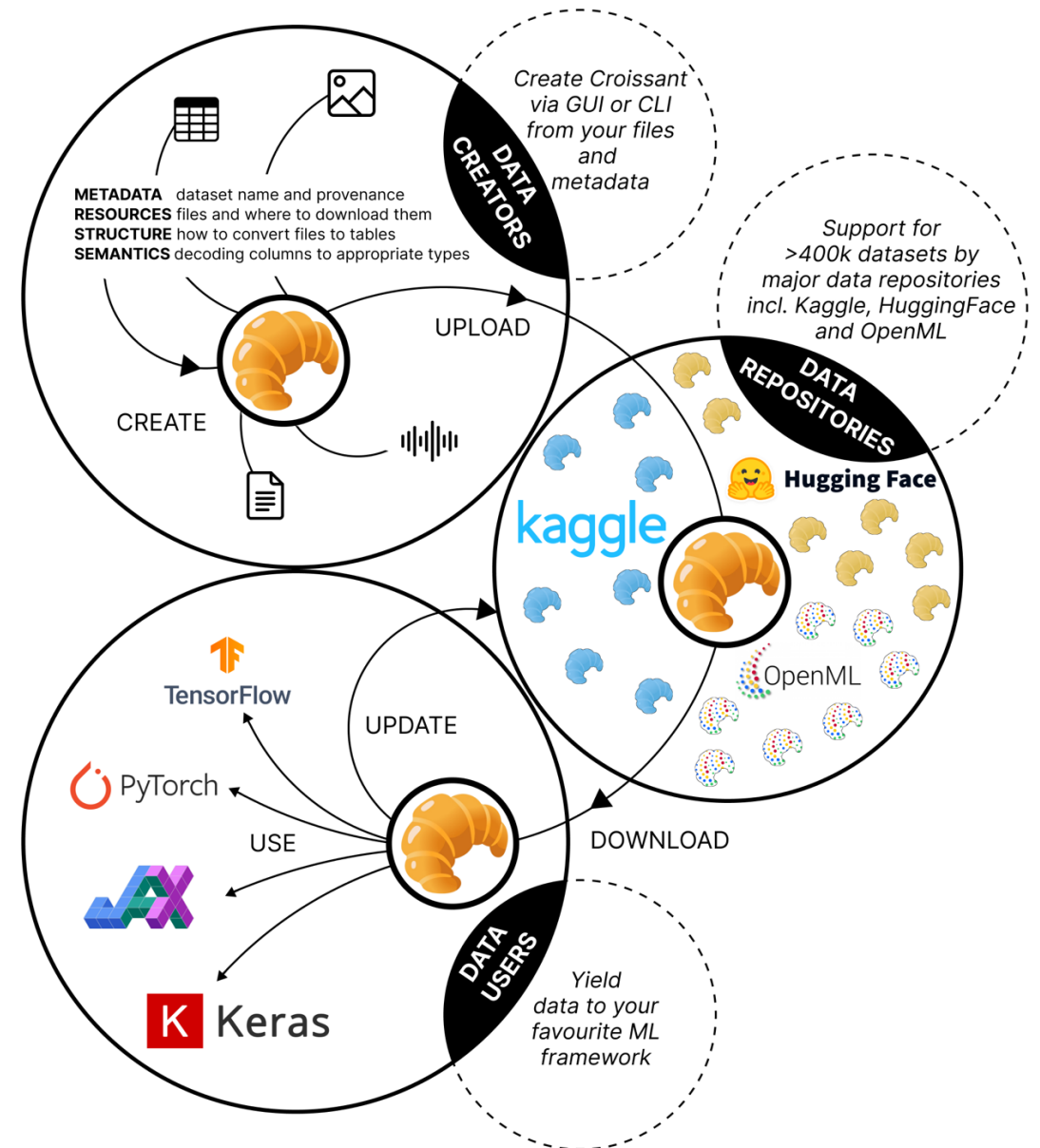
- **It describes everything AI needs to know about data**

- Access, structure, licensing, provenance, annotation, and intended use

Croissant: A Metadata Format for ML-Ready Datasets

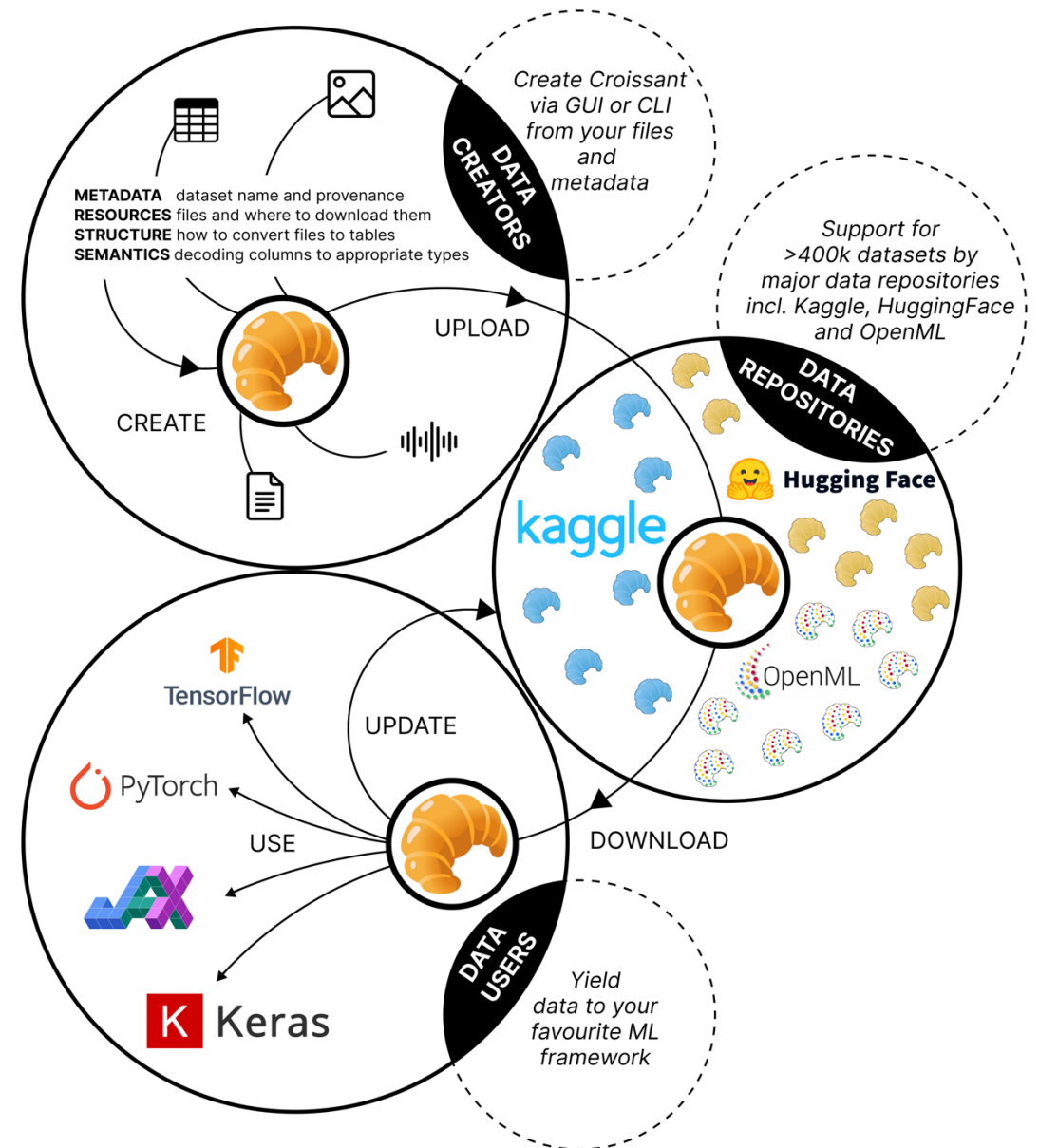
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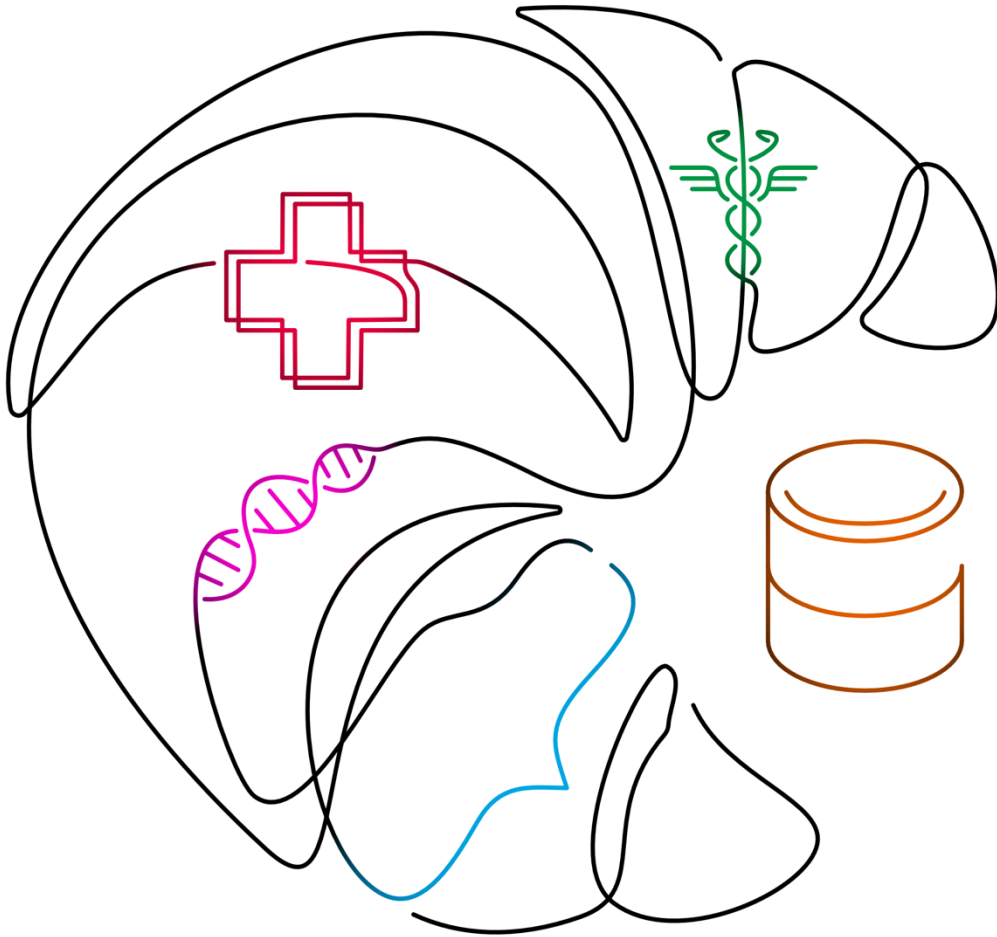


Data is Source Code for AI

- Croissant vocabulary does not require changing the underlying data representation
- Unified view over different layouts and formats
- Croissant adds structure and semantic
 - where is it stored? how to download?
 - how to cite? kind of license
 - how to read it? how to prepare the data?
 - who has annotated? annotation software used?
- Croissant file (json-type) follows schema.org syntax
 - Web-compatible
 - Machine readable & Searchable
 - Apache Beam support
- Responsible AI extension
 - Data Cards with harmonized vocabulary



Croissant on the inside



```
{
  "@type": "sc:Dataset",
  "name": "minimal_example_with_recommended_fields",
  "description": "This is a minimal example, including the required and the recommended fields.",
  "license": "https://creativecommons.org/licenses/by/4.0/",
  "url": "https://example.com/dataset/recipes/minimal-recommended",
  "distribution": [
    {
      "@type": "cr:FileObject",
      "@id": "minimal.csv",
      "name": "minimal.csv",
      "contentUrl": "data/minimal.csv",
      "encodingFormat": "text/csv",
      "sha256":
"48a7c257f3c90b2a3e529ddd2cca8f4f1bd8e49ed244ef53927649504ac55354"
    }
  ],
  "recordSet": [
    {
      "@type": "cr:RecordSet",
      "name": "examples",
      "description": "Records extracted from the example table, with their schema.",
      "field": [
        {
          "@type": "cr:Field",
          "name": "name",
          "description": "The first column contains the name.",
          "dataType": "sc:Text",
          "references": {
            "fileObject": { "@id": "minimal.csv" },
            "extract": {
              "column": "name"
            }
          }
        },
        {
          "@type": "cr:Field",
          "name": "age",
          "description": "The second column contains the age.",
          "dataType": "sc:Integer",
          "references": {
            "fileObject": { "@id": "minimal.csv" },
            "extract": {
              "column": "age"
            }
          }
        }
      ]
    }
  ]
}
```

Metadata



Structure



Resources



Semantics



How to use it?

Hugging Face Search models, datasets, users...

Datasets: Lab-Rasool/TCGA like 13 Follow Lab-Rasool 9

Modalities: Text Time-series Formats: parquet Languages: English Size: 100K - 1M ArXiv: arxiv:246

Tags: medical multimodal tcga oncology Libraries: Datasets pandas Croissant +1 License: cc-by-

Dataset card Data Studio Files and versions xet Community

About this format
Croissant is a standard format for dataset metadata. [Read more](#)

Download Croissant metadata
List datasets using Croissant

Dataset Viewer Auto-converted to Parquet </> API </> Er

Subset (5) clinical · 43.1k rows	Split (4) gatortron · 10.8k rows
-------------------------------------	-------------------------------------

```
pip install mlcroissant
```

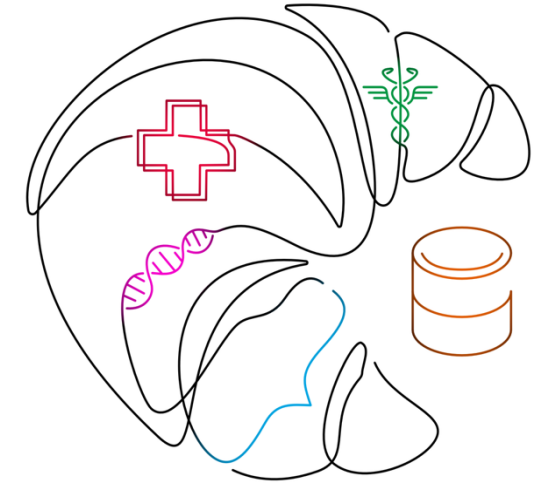
```
# 1. Point to a local or remote Croissant file
import mlcroissant as mlc
url = "https://huggingface.co/api/datasets/fashion_mnist/croissant"
# 2. Inspect metadata
print(mlc.Dataset(url).metadata.to_json())
# 3. Use Croissant dataset in your ML workload
import tensorflow_datasets as tfds
builder = tfds.core.dataset_builders.CroissantBuilder(
    jsonld=url,
    record_set_ids=["record_set_fashion_mnist"],
    file_format='array_record',
)
builder.download_and_prepare()
# 4. Split for training/testing
train, test = builder.as_data_source(split=['default[:80%]', 'default[80%:]'])
```

For practitioners – easy to use, just load your data from a single file (w/ context)

Why BioCroissant WG is needed – what AI for Health is missing

In analogy to the European Healthcare Data Space (EHDS):

Governance above; technical execution below



Non-normative & voluntary

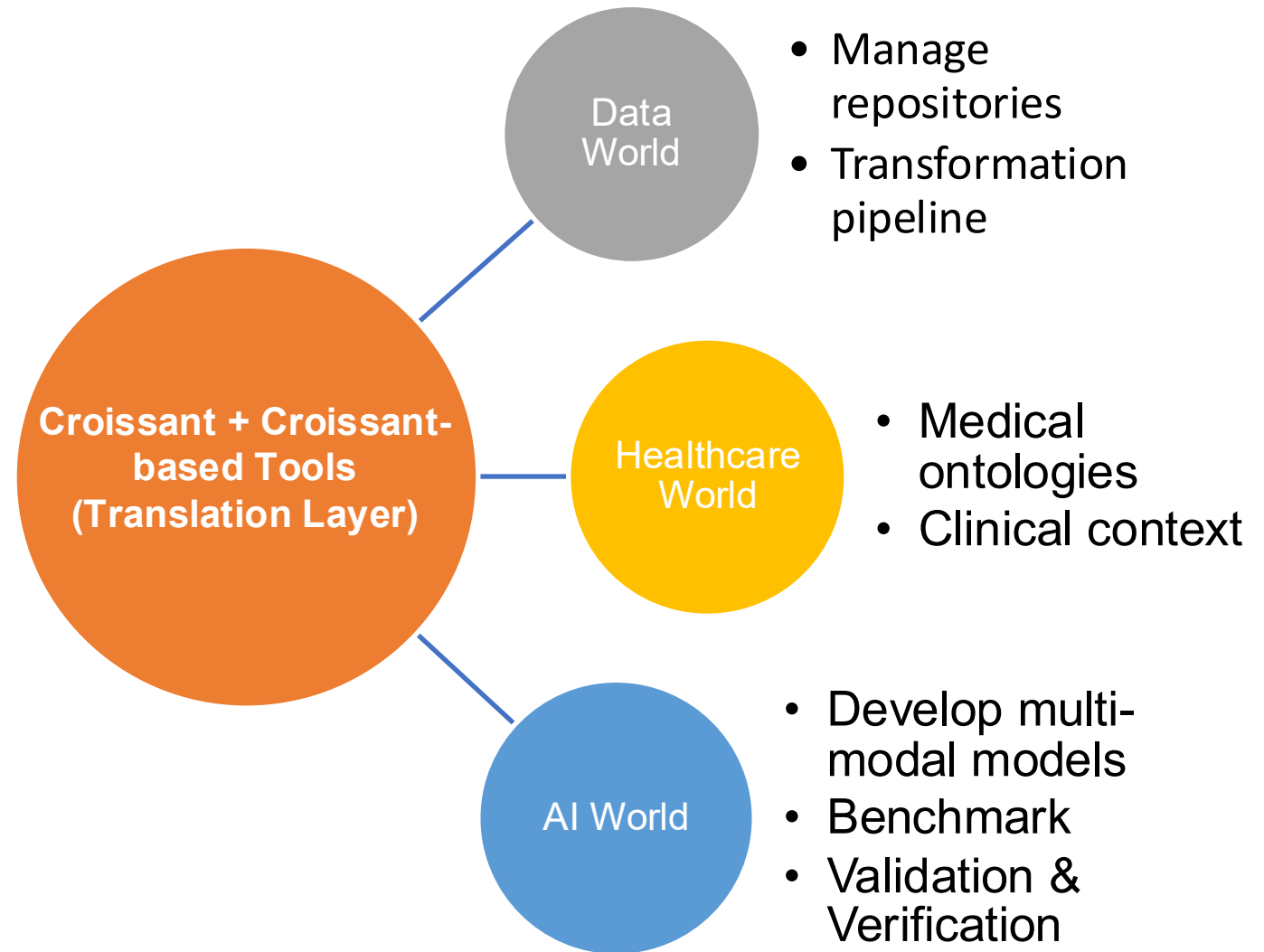
Downstream of EHDS governance

Enables auto-loading of datasets

EHDS-compatible technical profile

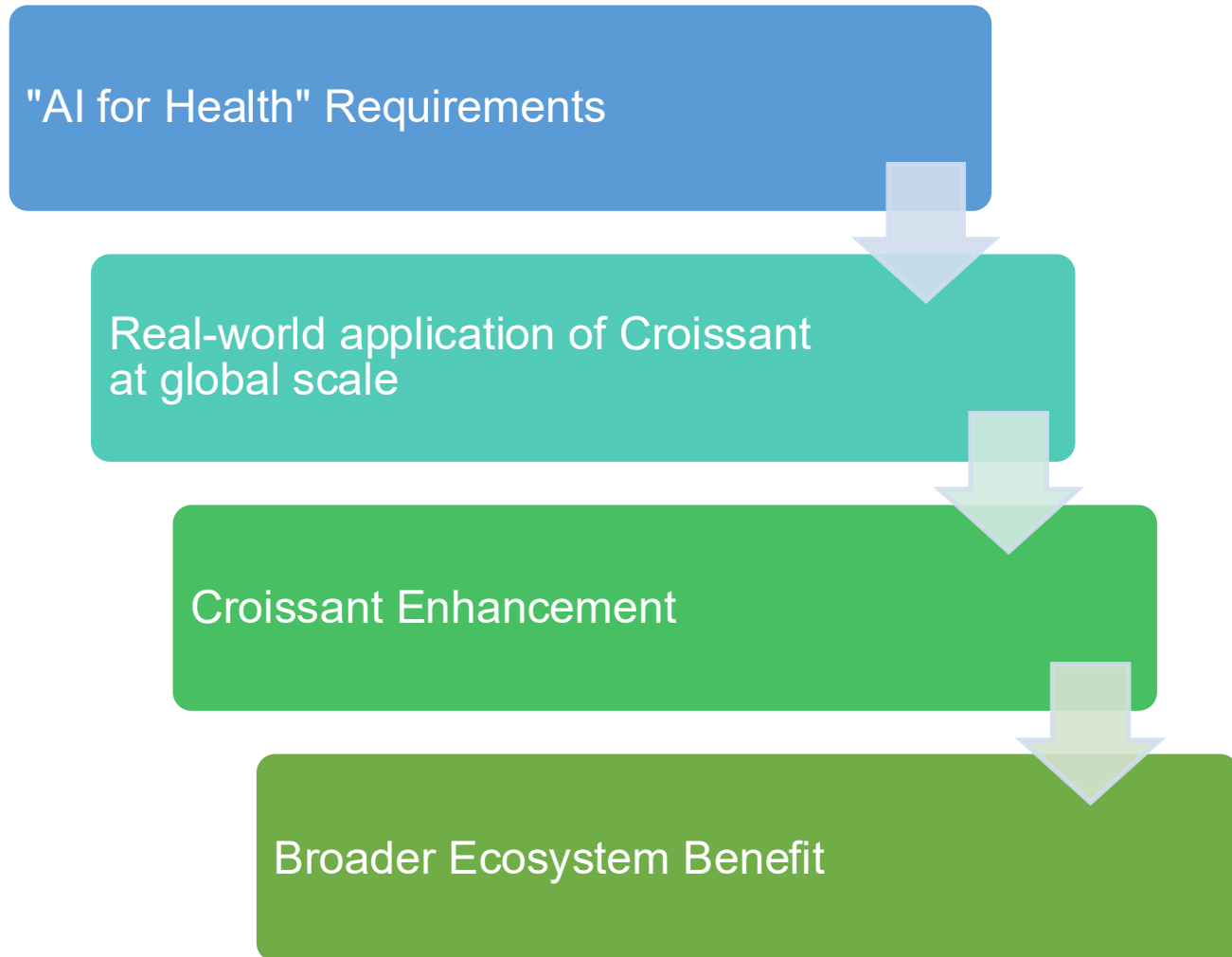
Goal: Bridging Three Communities with one data language

- Smart **metadata wrapping** across modalities, formats, and systems
- **Complexity stays under the hood.**
Tooling handles structure, provenance, licensing, and semantics
- **Experts stay in their lane.**
Each community works in familiar tools and concepts.

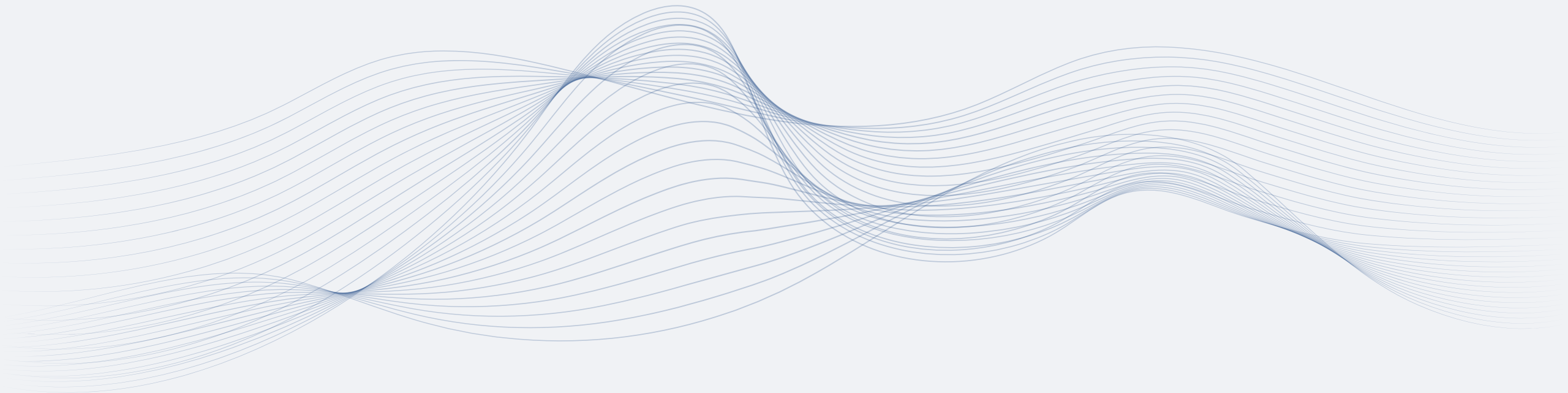


Selected to Build a Global Medical Data Catalogue

- Croissant has been selected by **ITU/WHO Global Initiative on AI for Health**
- "AI for Health" brings **real-world requirements** to Croissant
- Global benchmarking turns **theory into practice**
- **Feedback** flows back into Croissant and **improves** it



Building a Powerful Tool Ecosystem



BioCroissant Repository

BioCroissant — Metadata Sharing for (Bio)Medical AI

BioCroissant is a sub-working group of the [Croissant](#) initiative with strong ties to the MLCcommons [Medical working group](#) that is developing benchmarks and best practices. BioCroissant focuses on exploring and extending the Croissant standard for the biomedical domain, strengthening the technological link between AI and biomedical communities.

Its goal is to make medical and life-science data standards work seamlessly with the AI ecosystem—across ML frameworks, platforms, and Responsible AI tooling.

Built with real-world **clinical**, **research**, and **regulatory** needs in mind, BioCroissant bridges the gap between ML engineers, healthcare professionals, and policy makers working to build trustworthy health data ecosystems (e.g. academic health data consortia or international initiatives such as the European Health Data Space, EHDS).

Why BioCroissant?

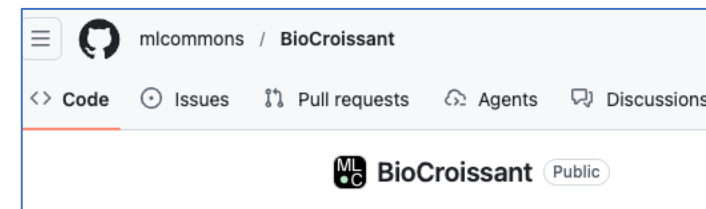
Modern biomedical AI depends on high-quality, interoperable data—yet many datasets today remain siloed, inconsistently documented, or difficult to reuse.

BioCroissant provides:

- **A simple, structured metadata schema** for biomedical datasets
- **Built-in FAIR alignment** (Findable, Accessible, Interoperable, Reusable)
- **Compatibility with national and cross-border health data initiatives**
- **Seamless integration with ML workflows** through clean, machine-readable metadata
- **Governance-friendly transparency**, supporting reproducibility, auditability, and clinical trust

Whether you are building models, conducting clinical research, or shaping data policy, BioCroissant helps ensure your data is **ready for science**, **ready for AI**, and **ready for regulation**.

<https://github.com/mlcommons/BioCroissant>



- Call for Sharing/Contribution/Feedback
- Roadmap is driven my discussion in the WG
- My favorite next topic:
 - Biomedical imaging formats
- FHIR support is around the corner

Discussions

-  2  **Dual-compatibility Croissant files following domain ontologies (e.g. OMOP, FHIR, and BioSchema profiles)**
[steffenvogler](#) started on Apr 28 in [Ideas](#)  2
-  1  **Data modalities beyond imaging**
[charliejharrison](#) started 3 weeks ago in [General](#)  2
-  5  **Support for biomedical imaging formats (WSI, microscopy, multi-resolution images, OCT etc)**
[steffenvogler](#) started on Apr 28 in [Ideas](#)  11
-  2  **Representing longitudinal patient-centric data**
[steffenvogler](#) started on Apr 28 in [Ideas](#)  0

Croissant Baker



Croissant Baker

Run Tests passing python 3.10+ uv License MIT pypi v0.3.0 PRs welcome

Automatically generate [Croissant](#) JSON-LD metadata for ML datasets — e.g. for [PhysioNet](#), [NeurIPS Datasets & Benchmarks](#) submissions, or any platform that benefits from standardized dataset metadata.

[Documentation](#)

Installation

```
pip install croissant-baker
```

or with [uv](#):

```
uv add croissant-baker
```

- CLI to auto-generate metadata files for local data
- Plays nicely with coding agents
 - Deterministic instead of guessing fields

Croissant Baker: Metadata Generation for Discoverable, Governable, and Reusable ML Datasets

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<https://github.com/MIT-LCP/croissant-baker>
<https://pypi.org/project/croissant-baker/>
<https://lcp.mit.edu/croissant-baker/>

- Manuscript has strong medical focus
- Submitted to NeurIPS “Evaluations and Datasets”
- <https://arxiv.org/abs/2605.15079>

<https://github.com/MIT-LCP/croissant-baker>

Eclair – Croissant’s MCP server solution

Search Extensions in ...

INSTALLED 21

RECOMMENDED 7

MCP SERVERS - INSTALLED

Eclair

Eclair

search for fashion datasets

I'll search for fashion datasets using the Eclair dataset search functionality.

Ran search-datasets

Joaquin Vanschoren

Find a fashion-related dataset and visualize some data examples

Great! I found several fashion-r

Core Fashion Dataset

Gemini

Thinking

Of course! I can help with that. First, I'll search for fashion-related datasets.

search-datasets (Eclair MCP Server)

fashion

You chose to always accept all local tools in this chat.

I found a few interesting datasets. Here are some of the most relevant ones:

- dataset/zalando-datasets/fashion_mnist: A dataset of Zalando's article images, consisting of a training set of 60,000 examples and a test set of 10,000 examples. Each example is a 28x28 grayscale image, associated with a label from 10 classes.
- dataset/Han-Xiao/Fashion-MNIST: A similar dataset to the one above, also with 60,000 training and 10,000 test examples.
- dataset/detection-datasets/fashionpedia: A dataset with images of fashion items, annotated with segmentation masks and their associated attributes.
- dataset/ashraq/fashion-product-images-small: A dataset of fashion product images.

The dataset/zalando-datasets/fashion_mnist and dataset/Han-Xiao/Fashion-MNIST datasets seem like good starting points as they are well-structured and ready for use. I'll proceed with dataset/zalando-datasets/fashion_mnist.

First, I'll get the dataset's metadata to understand its structure better.

dataset/microissant (Eclair MCP Server)

zalando-datas...

Thinking

The metadata confirms that this dataset contains images and corresponding labels, which is perfect for our task. Now, I'll download the dataset.

README.md

Eclair

Your lightning-fast data scientist

Eclair is a set of agentic tools that helps AI models (like Gemini and Claude) discover, download, and use datasets to answer questions based on factual data. Today, such models struggle to discover and use structured data to answer questions, which limits their scientific and analytical potential.

<https://github.com/mlcommons/croissant/tree/main/eclair>

```
graph LR; URL[https://github.com/mlcommons/croissant/tree/main/eclair] -- url --> FS[(File storage)];
```

Fashion Dataset Discovery and Visualization

This notebook demonstrates how to use Eclair to find fashion-related datasets and create compelling visualizations of the data examples. We'll explore the Fashion-MNIST dataset, which contains 70,000 grayscale images of fashion items across 10 different categories.

1. Import Required Libraries

First, let's import all the necessary libraries for data processing, visualization, and Eclair dataset access.

```
# Standard library imports
import asyncio
import json
import random
from collections import Counter

# Data processing and visualization
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from PIL import Image

# Hugging Face datasets (will be used based on Eclair's instructions)
from datasets import load_dataset

# Configure matplotlib for better plots
plt.style.use('default')
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['font.size'] = 10

print("All libraries imported successfully!")
```

> Find a fashion-related dataset and visualize some data examples

- I'll help you find and visualize a fashion-related dataset. Let me start by searching for fashion datasets.
- Update Todos
 - Search for fashion-related datasets
 - Analyze and select the best dataset
 - Download or access the dataset
 - Create visualizations of data examples
- Eclair - search-datasets (MCP)(query: "fashion")

2. Initialize Eclair Client

We'll set up the Eclair client to connect to the MCP server for dataset discovery and download.

Call to action: Join the Movement!

Integrate Croissant support into data repositories, analysis tools, and ML frameworks to expand ecosystem

Tool Developers



Reference Croissant standards in data governance frameworks to promote interoperability and FAIR compliance



Policy Makers



Researchers

Contribute to BioCroissant working group, share implementation experiences, and help refine health-specific extensions



Data Providers

Adopt Croissant for health datasets to improve discoverability and enable responsible reuse across platforms

